

CTDDG: Graph-Based Deep Generative Model for Low-Toxic Drug Design for a Target Protein with Optimal Binding Affinity

Nitesh Singh¹, Sahely Bhadra^{1,2*}, Pratiti Bhadra³

¹Mehta Family School of Data Science and Artificial Intelligence,
Department of Data Science, Indian Institute of Technology Palakkad,
Kanjikode, Palakkad, 678623, Kerala, India.

² Department of Computer Science and Engineering, Indian Institute of
Technology Palakkad, Kanjikode, Palakkad, 678623, Kerala, India.

³ Center for Computational Engineering and Networking, Amrita Vishwa
Vidapeetham, Amritanagar, Coimbatore, 641112, Tamilnadu, India.

*Corresponding author(s). E-mail(s): sahely@iitpkd.ac.in;

Contributing authors: sniteshreso22@gmail.com ;

b_pratiti@cb.amrita.edu;

Abstract

Designing diverse novel lead compounds with optimal pharmacological properties and low toxicity for specific disease-causing proteins remains a major challenge in drug discovery. Recent advancements in AI-based generative drug design have demonstrated promising potential for developing compounds with desired properties. Since toxicity is a critical factor, minimizing it is essential to ensure drug safety, therapeutic effectiveness, regulatory approval, and patient adherence. This study introduces CTDDG, a graph-based generative model that employs a deep-generative approach to design drug-like molecules with high binding affinity to target proteins while reducing toxicity. The model is trained to simultaneously maximize the likelihood of generating drug-like compounds and minimize the likelihood of toxicity. To the best of our knowledge, CTDDG is the first method designed to generate **less toxic de novo drugs for novel protein targets**. Approximately 95% of the molecules generated are valid and unique, with over 80% adhering to Lipinski’s rules. More than 95% of the generated compounds demonstrate favorable binding affinity with target proteins during testing. To demonstrate CTDDG’s application in drug discovery, we developed a molecular interaction-based virtual screening process that successfully identified

a novel Cyclin-Dependent Kinase 2 (CDK2) inhibitor, highlighting its potential to accelerate drug development.

Keywords: De-novo drug design, toxicity, graph neural networks, molecular interaction, virtual screening

1 Introduction

Developing new drugs for specific diseases is essential for advancing public health. The drug discovery pipeline generally involves several stages: target discovery, lead compound identification, synthesis pathway development, and lead optimization. On average, the entire process, from pre-clinical target identification to market approval, extends over a period of 10 to 15 years (Paul et al, 2010). This endeavor is both costly, often exceeding \$1.8 billion, and carries a significant risk of failure. Additionally, the drug discovery process is highly complex, requiring extensive experimentation across multiple stages, and is characterized by long timelines and substantial costs. Furthermore, navigating the vast chemical space, and discovering a molecule that balances multiple desirable properties, safety, and potency against a specific target, poses significant challenges (Koutroumpa et al, 2023). Traditional experimental methods face challenges such as limited efficacy, high costs, unintended off-target effects, and lengthy development times. Recently, computer-aided drug discovery has proven effective in managing extensive chemical data, consequently reducing both the time and costs associated with drug development (Vemula et al, 2023). Additionally, various AI-based methodologies have been integrated into different stages of the drug discovery process, significantly enhancing overall efficiency (Tripathi et al, 2023). Among these, generative models within artificial intelligence have attracted considerable attention as a promising approach to de novo drug design (Cheng et al, 2021; Tong et al, 2021).

In recent years, there have been groundbreaking advancements in the field of de novo molecular generation, which aims to create new molecules with specific properties. Various deep generative architectures, such as Variational autoencoder (VAE), generative adversarial networks (GAN), have been employed (Pang et al, 2023). **VAE-based Approaches:** VAE has been attractive in molecular generation in the virtue of its latent space is potentially operatable. CharVAE (Gómez-Bombarelli et al, 2018) first proposes to learn from molecular data in a data-driven manner and generate with a VAE model. By jointly training the task of predicting molecular properties from the hidden layers, this method explores the molecular space without manual prior knowledge. JT-VAE (Jin et al, 2018) first realize the direct generation of molecular graphs instead of linear SMILES (Simplified Molecular-Input Line-Entry System) strings. HierVAE (Jin et al, 2020) proposes a motif-based hierarchical encoder-decoder toward the large molecular graph generation. All the VAE methods mentioned earlier are designed to understand the distribution of a large set of drug-like molecules. This enables the generation of novel drug-like molecules that resemble those in the input set. **GAN-based Approaches:** An alternative is to implement

GAN in molecular generation. ORGAN (Guimaraes et al, 2018) adds expert-based rewards under the framework of WGAN (Arjovsky et al, 2017). MolGAN(Cao and Kipf, 2022) proposes GAN-based models to generate molecular graphs rather than SMILES. Mol-CycleGAN (Maziarka et al, 2020) generates optimized compounds with high structural similarity to the original ones. Both the models MolGAN and Mol-CycleGAN are designed to generate molecules which are similar to molecules which were present in training data and are also property optimized. **Flow-based Approaches:** Molecular generation with normalizing flow is promising, as its invertible mapping can reconstruct the data exactly. GraphNVP (Madhawa et al, 2019) defines a normalizing flow from a base distribution to molecular graph structures. MolFlow (Zang and Wang, 2020) proposes a variant of Glow (Kingma and Dhariwal, 2018) to generate atoms and bonds in a one-shot manner. MolGrow (Kuznetsov and Polykovskiy, 2021) constrains optimization of properties using latent variables of the model and recursively splits nodes to generate molecular structures.

However, none of these above stated methods take into account the scenario of protein-based drug generation, where drugs are designed specifically for a target protein. This limitation renders these methods directly inapplicable to target-based drug design specifically for those proteins having minimal drugs against it. **3D structure Approaches:** pocket2mol(Peng et al, 2022), 3D generative model(Luo et al, 2022) deal with the 3-dimensional structure of target and ligand. These models learn the distribution of atoms and bonds at the binding site of protein-ligand structure. Usually, 3D structure based generative models are only designed to generate drugs for known proteins as they require the pocket information of the target protein to model the distribution of atoms of the ligand.

Molecular lead optimization is a key step in drug discovery, where molecules similar to a drug candidate are developed with enhanced properties. Priority proposed generative models focused on optimizing general properties to produce drug-like molecules with better drug-likeness (QED), synthetic accessibility, logP (octanol-water partition coefficient), and molecular weight (Wang et al, 2023; Chen et al, 2023). However, unanticipated toxicities remain a major challenge in drug discovery and development, often preventing promising candidates from reaching the market (Kramer et al, 2007). Drug safety issues are responsible for the attrition of approximately 30% of drug candidates and can result in the costly withdrawal of many marketed drugs (Giri and Bader, 2015). Recently, a few studies have focused on generating new drug-like molecules with optimized toxicity profiles. Yang et al. developed a conditional generative model using a semi-supervised VAE to create molecules with low toxicity. They considered five toxicity factors: cardiotoxicity, mutagenicity, hepatotoxicity, pulmonary toxicity, and skin sensitivity. Their approach produced a higher proportion (approximately 25%) of non-toxic molecules compared to models focusing only on drug-likeness (Yang et al, 2022). Despite this, their model did not incorporate target-specific considerations or integrate toxicity into the molecular generation process. On the other hand, Kang et al. (Kang and Cho, 2019) focused on property optimization in their molecular generation process but did not address

target specificity or low toxicity. They introduced a model for conditional molecular design using SSVAE (Kingma et al, 2014), which generates molecules with desired properties by conditioning the distribution on these properties. This model efficiently generates new molecules that meet specified conditions on properties. However, it is not well-suited for designing drugs based on specific target proteins, as it does not explicitly incorporate target protein conditions. In contrast, DrugEx (Liu et al, 2021) incorporates target proteins and optimizes multiple properties, including toxicity, during molecule generation. DrugEx utilizes RNN-based reinforcement learning and multi-objective optimization to generate molecules targeting various proteins, although it is specifically designed for adenosine receptors. In our previous model, CDGCN used a sequential decoding scheme to generate protein-specific drugs for both known and novel proteins, incorporating protein information through pre-trained embeddings. Although CDGCN (Mallick and Bhadra, 2023) effectively generates drug candidates and learns the relationship between target proteins and ligands, its loss function prioritizes maximizing the likelihood of generating molecular graphs without explicitly optimizing for molecular properties like toxicity, binding affinity, QED, or SAS. Optimizing individual properties, such as toxicity, can negatively impact other crucial attributes like binding affinity and adherence to Lipinski’s Rule of Five. To address this, we updated CDGCN to simultaneously reduce toxicity and ensure strong binding affinity for novel target proteins, as proposed in this study.

In this study, we present Conditional Target-based Detoxed Drug Generation (CTDDG), a deep learning-based model for de novo drug design aimed at minimizing toxicity while ensuring strong binding affinity to a novel target protein. The model optimizes the generation of non-toxic molecules with drug-like properties and maintains a balance with other critical characteristics, such as binding affinity and adherence to Lipinski’s Rule of Five. The model employs a two-phase training process: first, it is pre-trained on toxic and non-toxic datasets to favor non-toxic molecule generation and avoid toxic pathways. In the second phase, it is fine-tuned to produce non-toxic, drug-like molecules with strong binding affinity. A novel loss function is introduced, comprising two components: one to minimize toxic molecular graphs and another to maximize non-toxic ones. This approach decreases the percentage of toxic molecules among the generated samples for novel proteins. To demonstrate CTDDG’s application in drug discovery, we conducted a case study to generate a new inhibitor for Cyclin-Dependent Kinase 2 (CDK2). We utilized molecular docking and simulations to identify the least toxic and most promising molecules. These molecules exhibited interaction patterns comparable to those of a well-known CDK2 inhibitor, suggesting that they may serve as potential drugs with similar molecular mechanisms.

2 Material and Methods

2.1 Datasets

The dataset for this experiment follows the setup of CDGCN (Mallick and Bhadra, 2023). Two real biological databases, i) ChEMBL (Mendez et al, 2018) and ii) BindingDB (Liu et al, 2006) were used for empirical evaluation. ChEMBL is a manually

curated public database of approximately 2.3 million molecules. 1093589 molecules satisfying the desired physicochemical properties (Section 3.4) were extracted and used for the pre-training of proposed method. The data set does not provide any information on the toxicity profile of the compounds. In our study, we focused on preventing the generation of molecules that can induce DNA mutations. DNA mutation is an alteration in the nucleotide sequence of an organism’s genome. These alterations have the potential to interfere with regular cellular processes, resulting in adverse outcomes such as cancer, genetic abnormalities, and cellular demise. Mutagenic agents, typically small molecules, induce mutations by causing damage to genetic material, leading to potential disruptions in normal cellular functions and contributing to toxicity. This necessity underscores the importance of evaluating Ames mutagenicity. Thus, toxicity is conceptualized within the framework of Ames mutagenicity(Wu et al, 2021).

BindingDB is a public database of experimentally measured binding energies between proteins and ligands (Gilson et al, 2016). All experiments here use the version of the BindingDB database constructed by Grechishnikova (Grechishnikova, 2021) where each protein is presented as FASTA sequences (Lipman and Pearson, 1985) and each ligand is presented as SMILES strings (Weininger, 1988). The entire drug dataset has 65 unique atom types and four unique bond types (single, double, triple, and aromatic). The final version of BindingDB dataset obtained after preprocessing mentioned in CDGCN (Mallick and Bhadra, 2023) contains 166400 pairs of proteins and ligands. The training set contains 1042 unique protein sequences and test set contains 104 unique proteins. The majority of protein sequences within the training set share less than 40% pairwise similarity between sequences, assessed via the Needleman-Wunsch global alignment algorithm provided by the EMBOSS package. Here, we also use the Ames test (Stead et al, 1981) to assess the mutagenicity of a ligand, considering its mutagenic potential to determine its toxicity. The similarity between protein sequences between train and test split is kept within 20% to make test set diverse enough to train and evaluate the models. The models were trained, validated and tested by the ligands present in the BindingDB dataset, hence to avoid data leakage during evaluation these ligands were removed from the modified ChEMBL dataset. From now on, we will refer to proteins that interact with known drugs as “known proteins” and proteins that do not interact with any known drugs as “novel proteins”. Proteins in the training data are considered “known proteins”, while proteins in the test data are considered “novel proteins”.

2.2 CTDDG:Conditonal Target based Detoxed Drug Generation

Architecture for CTDDG is based on that of CDGCN (Mallick and Bhadra, 2023). A drug molecule is represented as a graph $G = (V,E)$ where V is the set of nodes in G corresponding to the atoms in the drug molecule, and E is the set of edges in G corresponding to the bonds in the drug molecule. Each node has an atom type from a fixed set A of atom types and each edge has a bond type from a fixed set B of bond types. The proposed CTDDG model takes as input a protein as a sequence of amino acids and sequentially generates a molecular graph by incrementally adding a node or

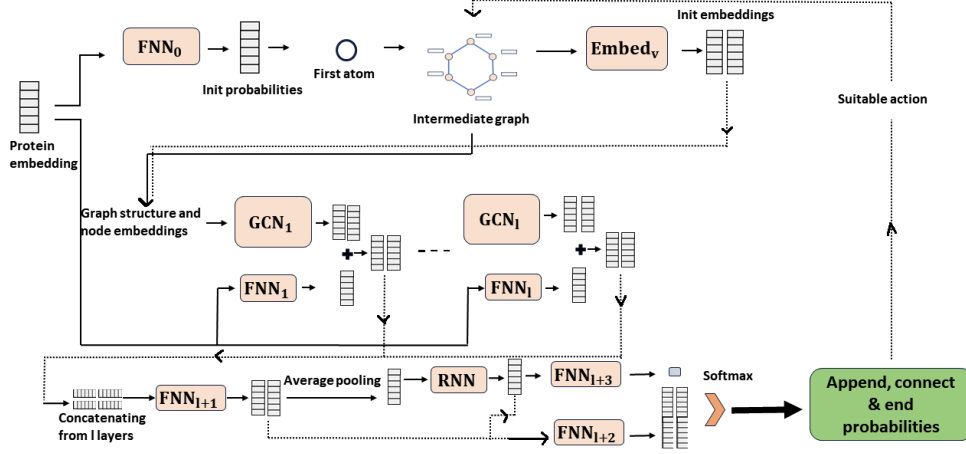


Fig. 1: Architecture of iterative process of the Conditional Target based Detoxed Drug Generation (CTDDG)

an edge. Figure 1 shows the graph generation process starts with an empty graph. At each generation step, CTDDG selects one of four different types of actions to be performed on the intermediate graph. i) *init* action that adds the first node to the empty graph, ii) *append* action that attaches a new node with a new edge to an existing node in the intermediate graph, iii) *connect* action which adds a new edge between an existing node and the last appended node in the intermediate graph, and iv) *end* action which ends the sequential graph generation process. Figure 1 shows an overview of the CTDDG architecture. CTDDG takes input a protein sequence transferred into embedding that contains useful information about its function. FNN_0 takes this protein embedding and gives probability scores for each atom type in A for the *init* action. At each generation step, $Embed_v$ takes the type of atom of all nodes in the intermediate graph and gives its embedding. GCN_1 takes the intermediate graph structure and the node embeddings and gives new node embeddings. The task-specific protein embedding generated by $FNN_1, \dots, FNN_l$ is summed with the embedding of the output node from $GCN_1, \dots, GCN_l$. Each of GCN_l modules takes the intermediate graph structure and the summed node embeddings from the previous layer and produces a new node embedding for the intermediate graph. The outputs of the l different summations are concatenated and passed to FNN_{l+1} to obtain the final node embeddings and then passed to an average pooling layer to obtain a graph-level embedding. This graph embedding is passed to RNN, which stores information from the previous intermediate graphs, and its output is concatenated with each node embedding from FNN_{l+1} . The concatenated outputs are passed to FNN_{l+2} to obtain a tensor of size $|V| \times ((|A| \times |B|) + |B|)$ for all possible append $|V| \times |A|$ i.e., with any existing node to any other atom and connect action i.e., add a bond with any node in graph and

to another existing atom $|V| \times (|A| \times |B|)$ or a new bond type with a node in graph $|V| \times |B|$. RNN output is passed to FNN_{l+3} to obtain a scalar for propability of the end action. Outputs from FNN_{l+2} and FNN_{l+3} are passed through softmax layer to obtain the final probability scores for the append, connect, and end actions. An action is then sampled to be applied on the intermediate graph. After append or connect action, the new graph is given as input to GCN_1 for the next generation step. After end action, the generation process stops and the current graph is post-processed to obtain the drug molecule.

2.3 Novel loss function to minimize likelihood of toxic samples

2.3.1 Initial loss function

The objective for CDGCN(Mallick and Bhadra, 2023) is to maximize the lower bound on the log-likelihood of generating a molecular graph G by following a generation path T given a protein P . T is a sequence of m actions where the j -th action t_j determines the evolution of the intermediate graph G_j at the j -th generation step. The next action depends on all the previous intermediate graphs and the protein P . The log-likelihood is given by:

$$\log(\mathcal{P}_\Theta(G, \mathcal{T} | P)) = \sum_{j=1}^m \log(\mathcal{P}_\Theta(t_{j+1} | G_j, \dots, G_1, P)) \quad (1)$$

where m is the number of actions in T and G_j is the intermediate graph obtained after applying the action $t_j \in T$. The marginal likelihood is given by

$$\log(\mathcal{P}_\Theta(G | P)) = \log \sum_{\mathcal{T} \in \mathcal{S}(G)} \mathcal{P}_\Theta(G, \mathcal{T} | P) \quad (2)$$

where $\mathcal{S}(G)$ is the set of all possible generation paths. However, for most molecules encountered during drug discovery, the marginal likelihood is intractable (Li et al, 2018). To resolve this issue, importance sampling is used to estimate a particular probability distribution by sampling from a different predefined distribution (Kloek and van Dijk, 1978) and (van Dijk and Kloek, 1983), and hence the marginal likelihood from Eq.2 comes down to

$$\begin{aligned} \log(\mathcal{P}_\Theta(G | P)) &= \log \mathbb{E}_{\mathcal{T} \sim Q_\alpha(\mathcal{T} | G, P)} \left[\frac{\mathcal{P}_\Theta(G, \mathcal{T} | P)}{Q_\alpha(\mathcal{T} | G, P)} \right] \\ &\geq \log \frac{1}{K} \sum_{k=1}^K \frac{\mathcal{P}_\Theta(G, \mathcal{T}_k | P)}{Q_\alpha(\mathcal{T}_k | G, P)} \end{aligned} \quad (3)$$

where $Q_\alpha(T | G, P)$ is a predetermined distribution over generation paths in which the actions at each step are either based on the ordering of canonical atoms (Weininger et al, 1989) with probability α or based on random ordering with probability $1 -$

α , and K denotes the number of different generation paths possible to generate G . The training of CDGCN (Mallick and Bhadra, 2023) is done in mini-batches of size N , and throughout the training process, the objective is to minimize the negative log-likelihood loss given by:

$$\hat{\mathcal{L}}(\Theta) = -\frac{1}{N} \sum_{i=1}^N \log \frac{1}{K} \sum_{k=1}^K \frac{\mathcal{P}_{\Theta}(G_i, \mathcal{T}_{i_k} | P_i)}{\mathcal{Q}_{\alpha}(\mathcal{T}_{i_k} | G_i, P_i)} \quad (4)$$

2.3.2 Predefined distribution: $Q_{\alpha}(T | G, P)$

(Li et al, 2018) has provided the algorithm for defining the predefined distribution over the various generation path of the ligands. Interestingly, this distribution gives you an edge on what is being learned. The parameter α (coefficient of randomness) which signifies the probability of choosing the next atom according to the canonical rank (Weininger et al, 1989) of the atoms among the neighbors of the last visited atom while traversing the generation path. This process is realized by a Bernoulli distribution which is parameterized by α . This gives you a kind of stochastic option to choose deterministic or a random path at each step of the traversal.

2.3.3 Novel loss function for CTDDG: Minimizing the likelihood of toxic samples

From Eq.4, we can observe that the loss maximizes the likelihood of generating graph G by following the path T . We want to avoid following the generation paths which results in toxic molecules. Hence a need to minimize the likelihood of path generating toxic molecules. Simultaneously, we have to maximize the likelihood of generating non toxic molecules. Here we try a loss function which is combination of minimization and maximization of likelihood. For simplicity, the loss is divided into two parts:

$$\max \hat{\mathcal{L}}(\Theta) = \frac{1}{N} \left(\lambda \hat{\mathcal{L}}_{\text{toxic likelihood}}(\Theta) + \hat{\mathcal{L}}_{\text{Non-toxic likelihood}}(\Theta) \right) \quad (5)$$

where $\lambda \geq 0$ is a trade-off parameter. Higher lambda indicates excessive suppression of toxic molecules which might limit the model’s exploration of certain important molecular scaffolds. Hence for our experiment, we have fixed λ to 1. The experiment shows that λ values produce safe but effective structures.

$$\hat{\mathcal{L}}_{\text{toxic likelihood}}(\Theta) = -\sum_{i=1}^N \left[\log \frac{1}{K} \sum_{k=1}^K \frac{\mathcal{P}_{\Theta}(G_i, \mathcal{T}_{i_k} | P_i)}{\mathcal{Q}_{\alpha}(\mathcal{T}_{i_k} | G_i, P_i)} \right] \mathbb{I}_{\{G_i \text{ is toxic}\}} \quad (6)$$

$$\hat{\mathcal{L}}_{\text{toxic likelihood}}(\Theta) = \sum_{i=1}^N \left[\log \frac{1}{K} \sum_{k=1}^K \frac{\mathcal{P}_{\Theta}(G_i, \mathcal{T}_{i_k} | P_i)}{\mathcal{Q}_{\alpha}(\mathcal{T}_{i_k} | G_i, P_i)} \right] \mathbb{I}_{\{G_i \text{ is non-toxic}\}} \quad (7)$$

Minimizing the subloss in Eq.6 is equivalent to reducing the likelihood of generating a toxic molecular graph for the target protein. Minimizing the subloss in Eq.7 is equivalent to maximizing the likelihood of generating non-toxic molecular graph for the particular target protein. Together, loss can be written as

$$\max \hat{\mathcal{L}}(\Theta) = \frac{1}{N} \sum_{i=1}^N \left[\log \frac{1}{K} \sum_{k=1}^K \frac{\mathcal{P}_{\Theta}(G_i, \mathcal{T}_{i_k} | P_i)}{\mathcal{Q}_{\alpha}(\mathcal{T}_{i_k} | G_i, P_i)} \right] \cdot (1 - (1 + \lambda) \cdot \text{Toxic_class}_i) \quad (8)$$

where Toxic_class_i is toxic class of i^{th} ligand. It is 1 for toxic ligand and 0 for non-toxic ligand. $P_{\theta}(\cdot)$ is distribution parameterized by a set of parameters θ that are realized by deep neural network. $Q_{\alpha}(\cdot)$ is predefined distribution over the generation paths parameterized by the user defined variable α . K is the number of generation paths considered, α is user defined parameter, N is the batch size, P_i is the i^{th} protein, and G_i is the molecular graph of ligand. In the given context, it becomes clear that for an individual data point, represented here by the ligand, only one aspect of the loss function is activated at a time, either the toxic or non-toxic component. If the ligand is predicted to be toxic, our objective is to minimize the likelihood of its generation; otherwise, it is to maximize the likelihood of its generation.

2.4 Training of CTDDG

BindingDB contains 66400 pairs of proteins and ligands pairs for 1146 unique protein sequences. This data is not enough to train the large architecture of CTDDG. Hence CTDDG get trained in two phases following Grechishnikova (2021) first phase is pre-training that trained CTDDG to generate valid non-toxic molecule. For pretraining, the Random Forest Classifier was used to classify ChEMBL molecules into toxic (Ames mutagenicity) and non-toxic groups. The training data for random forest is obtained from (Wu et al, 2021). During second phase of training CTDDG is trained to generate ligand specific to a target protein. For fine-tuning, BindingDB ligands were labelled using pkCSM to identify ligand with Ames mutagenicity

2.5 Molecular interaction and fingerprint

Docking: Smina(Koes et al, 2013a) was used for molecular docking to identify the top favorable generated molecules based on binding affinity. In this study, the docking box was set to 5 Å larger than the ligand, using the CDK2 binding pocket (PDB ID: 1AQ1) as the docking site, with default docking parameters; well-known inhibitors like Staurosporine, Dinaciclib, and Roscovitine were used as templates for initial virtual screening, with Roscovitine (Faber et al, 2023) selected for comparative molecular dynamics studies.

Simulation: The protein-ligand complex (CDK2 + generated inhibitor) models were placed in a water box using CHARMM-GUI (Lee et al, 2016), with a 150 mM NaCl solution added for electrical neutrality. MD simulations were conducted with GROMACS 2023.1 at 303.15 K using the CHARMM36m force field and TIP3P water model. The MMPBSA tool (Valdés-Tresanco et al, 2021) calculated binding free energy

and assessed hotspot residue contributions. MMPBSA combines molecular mechanics, Poisson-Boltzmann theory, and surface area calculations to estimate binding free energy. The Protein-Ligand Interaction Profiler (PLIP) (Salentin et al, 2015) was used to identify hydrogen bonds and hydrophobic interactions in the CDK2-inhibitor complex.

Fingerprint: To compare structural similarity between well-known inhibitors and generated molecules, we used MACCS key (Yin et al, 2019) and 2D pharmacophore fingerprints. The Molecular ACCess System (MACCS) uses a 166-bit fingerprint to indicate structural elements, while 2D pharmacophore fingerprints highlight key features for target interactions. Similarity between sets is measured by the Tanimoto coefficient (Chung et al, 2019), ranging from 0 (no similarity) to 1 (perfect similarity), calculated as $T(X_i, Y_i) = (X_i \cap Y_i) / (X_i \cup Y_i)$.

3 Results and Discussion

This section reports the quantitative evaluation of the proposed CTDDG. We compare the proposed method with the existing method in the following three aspects: 1) non-toxic ligand generation, 2) target awareness, and 3) drug-like properties of generated ligands.

3.1 Benchmarking methods

Our benchmarking methods are the following:

CDGCN: CDGCN (Mallick and Bhadra, 2023) is pretrained on ChemBL dataset with hyperparameters mentioned by the author on their github link. It is finetuned on BindingDB dataset with hyperparameters obtained mentioned by the author on their github link. It is tested on a test set containing 104 proteins.

SSVAE: Conditional molecular design based on SSVAE (Kang and Cho, 2019) is devoid of target aware but conditionally samples molecules of desired properties including toxicity. The CDGCN fine-tuning data set consisting of BindingDB ligands is decided to be the training dataset for SSVAE (Kang and Cho, 2019). BindingDB ligands are filtered based on tokens chosen in the default model of (Kang and Cho, 2019). For the supervised case, we have logP, QED and probability of toxicity predicted by our predictor as the set of properties so that later during inference, we can sample molecules conditionally on these properties. 10 sets of 100 molecules were sampled conditionally on toxicity probability 0.50.

Pocket2mol: For Pocket2mol (Peng et al, 2022), we have used the pretrained model provided by the authors on their github page. Then 10 sets of 100 samples are generated for each of the novel proteins.

3.2 Evaluation of proposed CTDDG

3.2.1 Drug like properties

Along with having a considerable binding affinity with the target protein, a lead should also have desirable physicochemical properties (Grechishnikova, 2021). Lipinski’s rule of five is a widely used rule of thumb to test whether the generated molecules have

Properties	CDGCN	SSVAE	CTDDG
Validity(%) $\uparrow$	91.7 ± 0.22	88.4 ± 3.29	94.4 ± 0.32
Uniqueness(%) $\uparrow$	91.7 ± 0.22	58.1 ± 4.13	94.4 ± 0.32
LogP < 5 $\uparrow$	68.6 ± 0.38	51.4 ± 4.31	74.8 ± 0.33
Molecular weight < 500 $\uparrow$	73.5 ± 0.47	51.4 ± 4.31	<u>71.1 ± 0.32</u>
No. of H bond donors < 5 $\uparrow$	<u>84.6 ± 0.25</u>	51.4 ± 4.31	87.9 ± 0.4
No. of H bond acceptors < 10 $\uparrow$	<u>84.5 ± 0.28</u>	51.4 ± 4.31	87.2 ± 0.43
No. of rotatable bonds < 10 $\uparrow$	80.3 ± 0.40	51.4 ± 4.31	<u>79.9 ± 0.38</u>
TPSA < 140 $\uparrow$	83.5 ± 0.23	51.4 ± 4.31	84.1 ± 0.36
QED(absolute) $\uparrow$	0.55 ± 0.00	0.44 ± 0.02	<u>0.51 ± 0.0</u>
SAS < 6 $\uparrow$	<u>84.7 ± 0.23</u>	51.4 ± 4.31	88.6 ± 0.37

Table 1: Drug-like properties of 100 generated samples in test set containing 104 test novel protein. **Bold** indicates best and underlined indicates second best result. The proposed model CTDDG is the best-performing model in almost all parameters.

certain physicochemical properties found in orally active drugs for humans like i) an octanol-water partition coefficient logP that does not exceed 5, ii) molecular weight within 500 daltons, iii) no more than 5 hydrogen bond donors (H-bond donors), and iv) no more than 10 hydrogen bond acceptors (H-bond acceptors) (Ghose et al, 1999). Molecules satisfying properties like i) no more than 10 rotatable bonds and ii) no more than 140 $\text{\AA}^2$ of topological polar surface area (TPSA) are predicted to have good oral bioavailability (Veber et al, 2002). The quantitative estimate of drug-likeness (QED) stands as another commonly employed property for the selection of suitable molecules in the initial phases of drug discovery. (Bickerton et al, 2012). It ranges from 0 to 1 with higher values indicating a highly drug-like molecule. The synthetic accessibility score (SAS) is crucial in deciding whether the molecule is easy to synthesize. (Ertl and Schuffenhauer, 2009a). We train all models with protein ligand pairs for 1042 proteins and tested with 104 proteins of BindingDB data following Mallick and Bhadra (2023). We present all the results in averaged percent of generated molecules for 104 test proteins satisfying that particular constraint over 10 runs in Table 1.

Inference Compared to the findings presented in (Mallick and Bhadra, 2023), our model retains and even improves certain properties, such as the valid percentage of molecules, uniqueness, and adherence to the Lipinski rule of five, as demonstrated in Table 1. This suggests that the drug-like properties of the generated samples are preserved and may even outperform previous results, indicating that toxicity optimization does not compromise these crucial properties.

3.2.2 Target awareness

To quantify the target awareness of the generated samples, we have reported the percentage of generated molecules exhibiting a binding affinity < -7.0 Kcal/mol with

PDB code of protein	CDGCN	Pocket2mol	SSVAE(No target awareness)	CTDDG
4nos(%) $\uparrow$	99.10 $\pm$ 1.08	72.49 $\pm$ 6.34	11.97 $\pm$ 3.04	95.55 $\pm$ 2.56
2nru(%) $\uparrow$	96.28 $\pm$ 2.15	<u>95.25 $\pm$ 2.91</u>	2.49 $\pm$ 1.19	93.33 $\pm$ 3.58
1lqf(%) $\uparrow$	<u>96.02 $\pm$ 1.51</u>	-	4.23 $\pm$ 1.38	98.03 $\pm$ 2.09
1k2r(%) $\uparrow$	98.57 $\pm$ 0.67	61.22 $\pm$ 11.1	19.04 $\pm$ 3.08	<u>95.71 $\pm$ 2.92</u>
1cqp(%) $\uparrow$	85.00 $\pm$ 4.07	97.74 $\pm$ 1.67	2.49 $\pm$ 1.19	<u>86.35 $\pm$ 3.43</u>
Avg(%) $\uparrow$	95.00 $\pm$ 1.90	81.68 $\pm$ 5.51	8.04 $\pm$ 1.98	<u>93.80 $\pm$ 2.92</u>

Table 2: Percentage of generated samples(set of 100) having binding affinity < -7.0 Kcal/mol with their target. $\uparrow$ shows higher is better. **Bold** indicates best and underlined indicates second best result. For 1lqf protein, Pocket2mol is not able to sample any molecule even after decreasing various thresholds for frontier atom and focal atom.

their respective target proteins. This percentage is presented as both the mean and the variance in 10 runs (Table 2). Given the computational intensity and time constraints associated with molecular docking, which hinder evaluation across entire test sets, we followed the approach outlined by (Mallick and Bhadra, 2023) and focused on same five novel proteins with available binding sites. We conducted molecular docking of their Protein Data Bank (PDB)(Berman et al, 2000) structures with the generated molecules and their known ligands using SMINA software(Koes et al, 2013b) with default settings

Inference In Table 2, we note the percentage of generated samples corresponding to five new proteins with a binding affinity less than -7.0 KCal/mol with their respective targets. This metric serves to quantify the model’s awareness of the target to some degree. CDGCN is the best model while the proposed CTDDG performs comparable to it on average. Pocket2mol performs low even though it exploits the information of binding site and PDB structure of the proteins. Its performance also varies greatly on different proteins. For SSVAE, we observe a relatively low percentage of generated samples meeting the criteria, indicating that SSVAE lacks target awareness. In contrast, our model’s performance is at least maintained, suggesting that the target awareness of the generated samples is preserved even during optimization for toxicity (Table 3).

3.2.3 Non-toxic ligand generation

To explore the toxicity profile of the generated samples, we are examining the proportion of toxic samples relative to the total number of generated samples for a specific protein target. Given that our research revolves around the generation of new drugs de novo, we have assessed the percentage of toxic samples for five proteins following the approach described by (Mallick and Bhadra, 2023). To determine whether a generated molecule is toxic or non-toxic, we utilized the freely accessible web server pkCSM, as provided by (Pires et al, 2015), for predicting ADMET properties of molecules. We have reported the averaged percentage for toxic molecules over 10 runs in Table 3,

PDB code of protein	CDGCN	Pocket2mol	SSVAE(No target awareness)	CTDDG
4nos(%) ↓	24.13 ± 4.52	<u>17.5 ± 3.77</u>	11.7 ± 4.13	19.84 ± 4.38
2nru(%) ↓	26.89 ± 5.31	30.1 ± 11.17	11.7 ± 4.13	23.32 ± 4.52
1lqf(%) ↓	23.12 ± 4.73	-	11.7 ± 4.13	<u>20.26 ± 3.88</u>
1k2r(%) ↓	22.49 ± 4.00	<u>17.9 ± 7.06</u>	11.7 ± 4.13	20.02 ± 7.29
1cqp(%) ↓	19.88 ± 4.11	36.2 ± 8.36	11.7 ± 4.13	13.17 ± 5.31
Avg(%) ↓	23.30 ± 4.53	25.43 ± 6.4	11.7 ± 4.13	<u>19.32 ± 5.08</u>

Table 3: Toxic percentage of generated molecules(set of 100 samples) for five novel proteins obtained from pkCSM. ↓ shows lower is better. **Bold** indicates best and underlined indicates second best result. For 1lqf protein, Pocket2mol is not able to sample any molecule even after decreasing various thresholds for frontier atom and focal atom.

believing that this metric adequately quantifies the model’s capability of generating non-toxic molecule for analysis purposes. This metric is reported for five novel proteins.

Inference Table 3 reveals a consistent decline in the proportion of toxic samples generated by our model when compared to CDGCN (Mallick and Bhadra, 2023). In contrast, pocket2mol (Peng et al, 2022), has a large variability in the percentage of toxic samples generated, indicating that pocket2mol may not inherently prioritize optimization for toxicity. According to average performance CTDDG is the best among method which can generate ligand for novel proteins. SSVAE produce very less toxic ligand but it could not produce good ligand with good binding affinity with novel protein (Table 2)

4 Case study: de novo design of Cyclin-Dependent Kinase 2 (CDK2) inhibitors

CDK2, also known as Cyclin-Dependent Kinase 2, is a serine/threonine protein kinase belonging to the cyclin-dependent kinase family. It regulates the cell cycle by promoting the transition from the G1 phase to the S phase, essential stages for DNA replication and cell division in normal cellular functions. Dysregulation of CDK2 is associated with various cancers, making it a significant target for cancer therapy and research. Recent studies have identified CDK2 as overexpressed in several human cancers, including colorectal, ovarian, breast, and prostate cancers (Tadesse et al, 2018). Dysregulation of CDK2 is associated with various cancers, making it a significant target for cancer therapy and research. Developing effective inhibitors against CDK2 holds promise in cancer treatment (Patel et al, 2023), particularly due to its critical role in cell cycle control and DNA processes, which are often disrupted in cancer cells. Hence, CDK2 has been chosen as a case study in this research to develop new inhibitors, utilizing CTDDG to illustrate its practical applications.

Name of molecules	Binding affinity ^a (Kcal/mol)	QED ^b	SAS ^c	Rat (LD50) ^d (mol/kg)	Flathead minnow toxicity ^e (mol/kg)	Lipinski violations
mol1	-13.09	0.28	6.45	2.72	2.05	2
mol2	-11.79	0.26	3.66	2.68	0.63	1
mol3	-11.73	0.36	3.64	2.92	2.77	0
mol4	-11.66	0.53	4.26	2.40	0.55	0
mol5	-11.65	0.22	3.78	2.52	-1.0	1
mol6	-11.59	0.30	4.2	2.58	-0.92	1
mol7	-11.52	0.67	3.97	3.39	0.87	0
mol8	-11.46	0.60	3.82	2.30	1.23	0
mol9	-11.130	0.40	3.77	2.40	0.626	1
mol10	-11.26	0.40	3.21	2.9	0.02	0
Staurosporine	–	0.40	4.93	2.39	3.16	0
Dinaciclib	–	0.47	4.02	2.58	2.55	0
Roscovitine	–	0.58	3.58	2.2	3.16	0

Table 4: Metrics related to toxicity, drug-likeness, and synthetic potential of the top ten favorable generated CDK2 inhibitors using CTDDG. Staurosporin, Dinaciclib and Roscovitine are well-know CDK2 inhibitors **Bold** indicates the best molecules based on toxicity.

^acalculated using SMINA(Koes et al, 2013a). ^bQED: Quantitative Estimate of Drug-likeness (Mazuz et al, 2023). ^cSAS: Synthetic Accessibility Score (Ertl and Schuffenhauer, 2009b). ^dLethal dosage values (LD50) are standard measures of acute toxicity used to compare the toxicity of different substances. LD50 represents the amount of a compound given at once that causes death in 50% of a group of test rats. (used pkCSM webserver)(Pires et al, 2015). ^eLethal concentration values (LC50) represent the concentration of a molecule required to cause the death of 50% of Flathead Minnows, a species commonly used in aquatic toxicology studies. (used pkCSM webserver)(Pires et al, 2015)

Binding affinity: Molecular docking was conducted to identify the top 10 most favorable inhibitors of CDK2 from a set of 100 inhibitors generated using CTDDG. The top 10 most favorable refers to the inhibitors with the lowest binding affinity (see Table 4). The FASTA sequence of human CDK2 (Uniport ID: P24941) was utilized to generate inhibitors using the CTDDG method. Supplementary Table S1 displays the SMILES notation for the top 10 most favorable inhibitors generated using CTDDG, alongside a well-known CDK2 inhibitor. We calculated the QED, SAS (Synthetic Accessibility Score), Rat LD50 (toxicity) and LC50 of Flathead Minnows (toxicity) metrics from the top 10 generated molecules and well-know inhibitor (see Table 4).

Drug-likeness and toxicity measure:The Synthetic Accessibility Score (SAS) estimates how easy it is to synthesize a molecule. The SAS ranges from 0 to 10, with values above 6 indicating a molecule is challenging to synthesize. Except for mol1, all generated molecules have an SAS below 4.5. The QED values of the well-known inhibitors range from 0.4 to 0.58, and the QED values of the five generated molecules also fall within this range. Notably, mol7 has the highest QED value at 0.6. Table 4 shows that the toxicity levels of the generated molecules are similar to those of

the well-known inhibitors. However, mol7 exhibits the lowest toxicity among the generated molecules based on the Rat LD50 metric. For the Fathead Minnow toxicity LC50 metric, only mol3 shows similar toxicity levels to the well-known inhibitors, while others exhibit higher toxicity.

Interaction patterns: The number of Lipinski’s rule violations (Nogara et al, 2015) was used as a criterion for the second stage of screening, following binding energy screening. Similar to the well-known inhibitors (Staurosporine, Dinaciclib, Roscovitine), the generated molecules mol3, mol4, mol6, mol8, and mol10 did not violate any of Lipinski’s rules. Consequently, these five generated inhibitors were selected for further analysis of their interaction patterns with CDK2 (see Table 5). PLIP (Salentin et al, 2015) was employed to generate the amino acid interaction fingerprints. It was observed that all well-known inhibitors (S,D,R) form a hydrogen bond with LEU83 (see Table 5). Previous research has also documented that the LEU83 interaction is present in 66 entries of CDK2 crystal structures with inhibitors (Mohammad et al, 2019; Liang et al, 2020). Most inhibitors occupy the purine binding pocket and form 2 or 3 hydrogen bonds with GLU81 and LEU83. In a similar fashion, as shown in supplementary Figure S1, the generated molecules mol3 and mol7 also form a hydrogen bond with LEU83. Therefore, we conducted MD simulations of CDK2-inhibitor using these two inhibitors (mol3 and mol7) to investigate their stability within the binding pocket and the stability of their interaction with LEU83 throughout the simulation trajectory. Figure 2 displays the docking poses of inhibitors (mol3 and mol7) obtained from the SIMNA docking simulation, which are used as the starting structures for the MD simulation.

Binding energy calculation through MM-PBSA from molecular dynamics simulation trajectories: The three protein-ligand systems are shown in supplementary Figure S2. The three systems are CDK2-Roscovitin, CDK2-mol3 and CDK2-mol7 complexes. Roscovitin (also known as CY-202 or Seliciclib) is a small molecule inhibitor of cyclin-dependent kinase 2, with an IC50 of 0.7 M. Its effects have been studied across a wide range of cancer cell lines (Cicenas et al, 2015). Therefore, Roscovitin is used for comparative analysis in MD simulations alongside mol3 and mol7. Analysis of the final conformations obtained from the MD simulation trajectories reveals that the interaction patterns of mol3 and Roscovitin are similar (supplementary Table S2). Both molecules form hydrogen bonds with LEU83 and exhibit comparable hydrophobic interactions with ILE10, VAL18, ALA31, and PHE80, suggesting that they occupy the same binding pocket. In contrast, the hydrogen bond between mol7 and LEU83 is disrupted. The binding energy between CDK2 and its inhibitors was calculated for the last 5 ns (50 frames) of the MD trajectory using the Molecular Mechanics Poisson–Boltzmann Surface Area (MM-PBSA) method. The binding energies for mol3, mol7, and Roscovitin are -12.05, -6.28, and -13.42 kcal/mol, respectively, suggesting that mol3 may be a potential inhibitor for CDK2. Furthermore, the per-residue decomposition analysis shows that the energy contributions of ILE10, PHE82, and LEU83 are strongly involved in the binding of both mol3 and Roscovitin. Conversely, ASP145 contributes less with a positive

Name of molecules	Hydrogen Bonds	Hydrophobic Interactions
mol3	ILE10, LEU83 , ASP86	ILE10, LYS33, PHE80, PHE82, LEU134
mol4	GLY13	ILE10, VAL18, ALA31, LYS33, PHE80, PHE82, GLN131, ASN132, LEU134, ALA144, ASP145
mol7	LEU83 , LYS89	ILE10, VAL18, ALA31, VAL64, PHE80, LEU134
mol8	HIS84	ILE10, VAL18, LYS33, VAL64, PHE80, PHE82, ASN132, LEU134
mol10	LYS33, ASP145	ILE10, VAL18, ALA31, VAL64, ASN132, LEU134, ALA144
Staurosporin	LEU83 , GLN131	ILE10, VAL18, ALA31, LYS33, LEU134, ASP145
Dinaciclib	LYS33, LEU83 , LYS89	ILE10, VAL18, ALA31, VAL64, PHE80, PHE82, GLN131, LEU134, ALA144
Roscovitine	GLU12, LEU83	ILE10, VAL18, LYS33, PHE80, PHE82, ASN132, LEU134, ALA144

Table 5: Hydrogen bonds and hydrophobic interactions between inhibitors and CDK2.

binding energy (see Figures 3(A) and 3(B)). In contrast, the energy contributions of residues in the binding of mol7 (see Figure 3(C)) differ completely from those of mol3 and Roscovitine. The contribution of residues to the binding energy of the inhibitor throughout the last 5 ns of the simulation is shown in supplementary Figure S3.

Diversity of generated molecules (mol3) from Roscovitine: The Tanimoto coefficients based on MACCS keys and 2D pharmacophore fingerprints between mol3 and Roscovitine are 0.46 and 0.21, respectively, indicating that mol3 exhibits diverse structure and chemical characteristics compared to Roscovitine. The QED score of mol3 and Roscovitine differs by approximately 0.2. (see Table 4). Therefore, new inhibitors like mol3, which possess different structures and chemical properties yet exhibit similar interactions, may provide better selectivity and efficacy. Supplementary Figure S2 shows the 2D structure of both molecules.

5 Conclusion

In this work, the first deep conditional graph generative network, CTDDG, was introduced for the generation of novel non-toxic but protein-specific de novo drugs. In addition to developing the methodology, we have demonstrated the use of CTDDG to discover novel inhibitors for any target protein and proposed a screening pipeline to identify lead molecules from generative models using biophysical insights. This approach identified a novel potential CDK2 (Cyclin-Dependent Kinase) inhibitor, with interaction patterns and binding residue contributions similar to roscovitine, a drug in

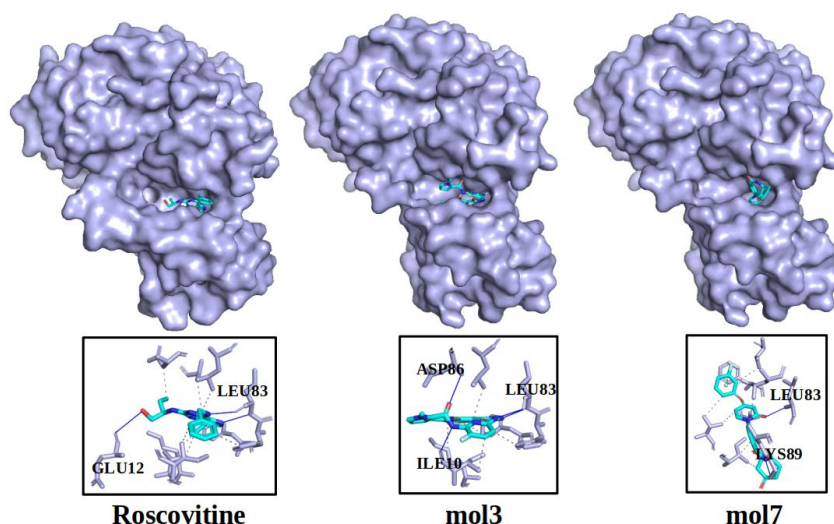


Fig. 2: Roscovitine (PB-ID: 2A4L), mol3 (middle), and mol7 (right) were docked to the experimental Cyclin-Dependent Kinase 2 structure. The protein is shown in a surface model (grey color), while the inhibitors are represented in a bond model. Lower-panel: blue and dotted lines represent hydrogen bond and hydrophobic interaction

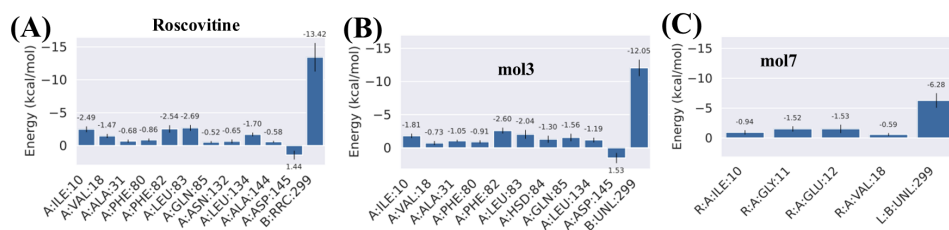


Fig. 3: (A-C) Residues that contributed the critical binding energies to the inhibitor during the per-residue MM-PBSA decomposition analysis and ΔG_{MMPBSA} of inhibitor, calculated using gmxMMPBSA tool from MD simulation trajectories

clinical trials. However, experimental validation through wet-lab studies is a next crucial step to demonstrate the real-life benefits of the model. At present, this is beyond the scope of the current manuscript. As part of our future work, the graph encoding scheme of CTDDG will be improved upon by exploring more efficient methods that provide more informative representations of molecular graphs, such as maximizing binding affinity, optimizing drug like proterties etc.

Supplementary information. **Supplementary.pdf:** Table S1: SMILES of generated and well-known inhibitors of CDK2; Table S2: Hydrogen bonds and hydrophobic interactions between inhibitors (mol3, mol7 and Rpscovitine) and CDK2 in the final

frame of the 50 ns MD simulation. Figure S1: Initial conformations of the molecular dynamics simulations. Figure S2: Per-residue binding energy decomposition analysis of Roscovitine (upper panel) and mol3 (lower panel) using the MM-PBSA approach from the last 5 ns of the Molecular Dynamics Simulation trajectory. Figure S3: 2D structure of mol3 and Rosocovitine.

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6 Declaration

6.1 Author Contribution:

Conceptualization and Supervision: P.B and S.B; AI-Methodology and Development: N.S. and S.B.; Computational Biophysical Investigation: N.S., P.B. and S.B; Writing Draft: N.S, P.B and S.B ;

6.2 Data availability:

Generated molecules and code are available at <https://github.com/sahelybhadra/CTDDG.git>. The docking structures of the top 10 most favorable generated molecules with CDK2 are located in the folder named 'Molecular_docking_top10GeneratedMol.' Additionally, the initial and final conformations from the Molecular Dynamics trajectories can be found in the "Molecular_Dynamics_files" folder. The data to train the model are available at this link : <https://drive.google.com/drive/folders/1KnQd6NFCiNNE6X1PvCryp4McDicBXp?usp=sharing>.

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